

Following information is required for drafting of patent application.

1. Full name, nationality and address of applicant(s):

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2. Full name (including middle name), nationality, address, mail id, and phone number of inventor(s):

Full Name (Including middle name)	Nationality	VIT Address (Start with full dept. name followed by full institute name) (For ex.)	Mail ID	Phone No.
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Copy and paste clear soft copy of signatures of all inventors in the following table: (Insert or delete the cells as per number of inventors)

Soham Moholkar	Shashank Kakad	Omkar Kakade	Yash Pardeshi

Rupali Mahajan			

3. Title of the invention:

It must describe your invention in one sentence. (Not more than 15 words)

Hybrid Soft Computing System for Uncertainty-Aware Knowledge Graph Construction and Explainable Retrieval

4. Technical field of the invention:

Ex. which streams of science or technology your invention is related to.

The invention relates to computer engineering, knowledge graphs, information retrieval, fuzzy logic, evolutionary computation, swarm intelligence, and explainable artificial intelligence.

5. Prior art:

What is the existing technology available in the field relating to your invention and disadvantages in that technology which are overcome by your invention. You can take the applications found in prior art search and mention how your invention is advantageous than the mentioned application.

Existing knowledge-graph and retrieval systems commonly use hard confidence thresholds, static ranking weights, vector-only retrieval, or conventional graph traversal. These approaches may discard uncertain but useful relations, preserve noisy edges, require manual parameter tuning, or provide limited traceability. The proposed invention integrates fuzzy confidence, adaptive optimization, confidence-aware multi-hop graph search, semantic retrieval, and source provenance in one system. Specific patent and publication references shall be added after a formal prior-art search.

6. Objective(s) of Invention:

Reason and advantages of your invention over existing technology

To construct confidence-weighted knowledge graphs from uncertain source material; adapt retrieval and graph-search parameters using evolutionary or swarm optimization; discover useful multi-hop graph paths using ant-colony search; and return results linked to source evidence, graph paths, and confidence information.

7. Synopsis:

Brief summary of your invention (if it is a device what the device is? Components required for construction of the device. How the components are assembled for construction of the device)

The system receives electronic source material, extracts entities and relations, assigns fuzzy confidence values, resolves equivalent entities, and stores confidence-weighted relations with provenance in a knowledge graph. For a query, semantic and graph retrieval are combined; ant-colony search explores

multi-hop paths; evolutionary or swarm optimization tunes retrieval parameters; and the selected result is returned with supporting evidence and graph-path information.

8. Brief description of drawings (if any):

If any figures are given to describe the invention only numerals must be used on it and what numerals denote must be described here. (As per new amendment in patent law, please use common single numerals for denoting any specific component in every figure)

Figure 1 illustrates the overall architecture of the proposed system. Figure 2 illustrates knowledge-graph construction and adaptive feedback. Figure 3 illustrates query processing, graph-path exploration, ranking, and evidence generation.

110 denotes data-ingestion module; 120 preprocessing/chunking module; 130 entity and relation extraction module; 140 fuzzy uncertainty engine; 150 entity-resolution module; 160 knowledge-graph store; 161-165 graph nodes; 170 semantic vector store; 180 optimization engine; 190 ant-colony path-search module; 200 hybrid retrieval/ranking engine; 210 provenance/explainability engine; 220 query interface; 230 result output interface; 240 evaluation/feedback module; 290 parameter repository; 300 source-document repository; and 310 evidence package.

9. Detailed description of the invention:

Describe in detail about the invention under this heading including any experimentation work carried out or validation study.

Source material is ingested, segmented, and processed to extract candidate entities, relations, and supporting evidence. A fuzzy inference engine combines uncertainty signals such as extraction confidence, semantic similarity, source reliability, or graph consistency to assign a relation-confidence value. Entity resolution merges equivalent entities and the resulting nodes, typed relations, confidence values, and provenance are stored in a knowledge graph. During retrieval, semantic candidates and graph candidates are combined. An ant-colony module explores relevant multi-hop paths using edge-confidence and pheromone information, while an evolutionary or swarm optimizer tunes ranking weights, thresholds, path penalties, or related parameters. A hybrid ranker selects evidence and an explainability module returns the source location, graph path, and confidence information. No unverified numerical performance claim is made in this proposal.

10. Best method of performance of the invention:

Preferably in one to two paragraphs mention the process in which invention is performed by giving example. (Actual practical working steps of your invention)

In a preferred embodiment, technical documents are split into source-linked units, entities and relations are extracted, fuzzy confidence values are assigned, duplicate entities are resolved, and a weighted knowledge graph is stored. On receiving a query, semantic retrieval identifies relevant evidence, ant-colony search explores graph paths, and an optimized hybrid ranker selects the strongest evidence. The output includes the source passage, selected graph path, and confidence information; evaluation feedback may subsequently be used to retune search parameters.

11. CLAIMS:

Mention specific features of your invention which you need to protect.

1. A computer-implemented knowledge processing system comprising a data-ingestion module, an entity-and-relation extraction module, a fuzzy uncertainty engine that assigns relation-confidence values, an entity-resolution module, a knowledge-graph store retaining source provenance, an ant-colony multi-hop path-search module, an evolutionary or swarm optimization engine, a hybrid semantic-and-graph ranking engine, and an explainability engine that outputs a result with supporting source evidence and graph-path

information.

2. The system of claim 1, wherein the fuzzy uncertainty engine combines at least two uncertainty signals selected from extraction confidence, semantic similarity, source reliability, cross-document support, type compatibility, evidence density, and graph consistency, and uses the resulting relation-confidence value to control insertion, weighting, down-weighting, or reevaluation of a graph relation.

3. The system of claim 1, wherein the optimization engine tunes one or more ranking weights, thresholds, path penalties, fuzzy parameters, or pheromone parameters using a genetic algorithm, particle swarm optimization, grey wolf optimization, whale optimization, or a hybrid thereof, and wherein ant-colony search uses confidence-aware graph information to identify multi-hop evidence paths.

4. A computer-implemented method comprising receiving and segmenting source material while preserving provenance; extracting candidate entities and relations; generating fuzzy relation-confidence values; resolving entities and storing confidence-weighted relations in a knowledge graph; receiving a query; obtaining semantic and graph candidate evidence; exploring multi-hop paths using ant-colony search; ranking candidate evidence using parameters determined at least in part by evolutionary or swarm optimization; and outputting a ranked result with supporting source evidence, graph-path information, and confidence information.

12. Inventive step of your invention:

Highlight what are the innovative features of your invention which makes your invention different than present technology. (For patentability, your invention must achieve either technical advantage over existing technology or it must reduce the cost of existing technology. Mention **HOW** this advantage is achieved by your invention)

The inventive step is the coordinated use of fuzzy confidence during graph construction, confidence-aware ant-colony multi-hop search, evolutionary or swarm parameter optimization, and provenance-linked explainable retrieval. The same uncertainty information that influences graph-edge weighting also influences path exploration and final ranking, thereby reducing dependence on rigid thresholds and fixed manually chosen parameters while preserving traceability to source evidence.

13. Industrial application:

In which field or purpose your invention is useful?

The invention is useful in enterprise knowledge management, research and education platforms, technical-document search, compliance and policy navigation, scientific literature exploration, software documentation, cybersecurity knowledge systems, and other applications requiring retrieval from uncertain or heterogeneous information.

14. Abstract:

Basic idea about your invention in one paragraph

A computer-implemented system constructs and queries an uncertainty-aware knowledge graph from electronic source material. Extracted entities and relations are assigned fuzzy confidence values and stored with source provenance. Semantic retrieval is combined with graph retrieval, ant-colony search explores multi-hop paths, and evolutionary or swarm optimization adjusts retrieval parameters. A hybrid ranking engine selects relevant evidence, and an explainability engine returns the result together with supporting source evidence, graph-path information, and confidence values.

15. Drawing

Drawings must not be labeled anywhere. Only numbers must be used to denote any

part and what the numbers denote is to be mentioned in brief description of drawings.

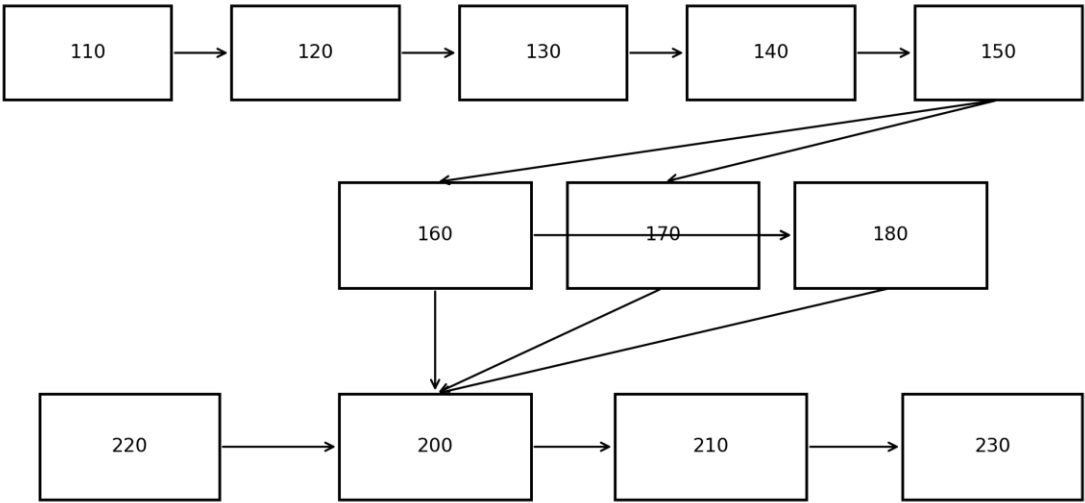


Figure 1

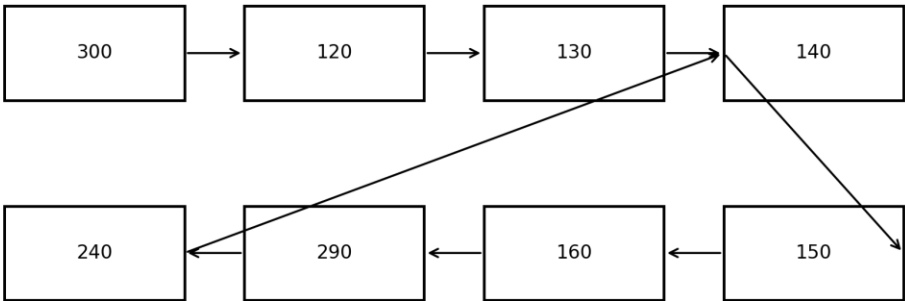


Figure 2

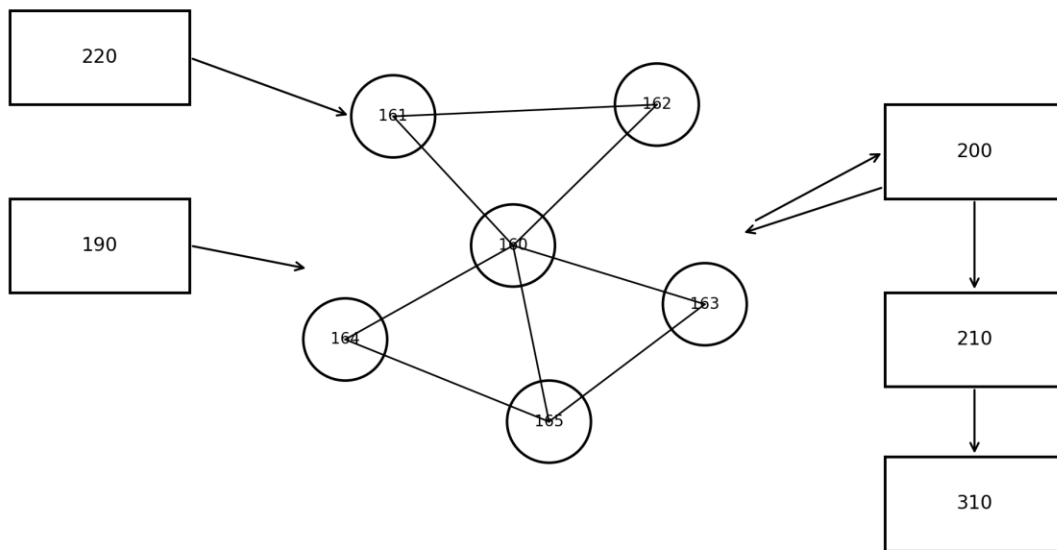


Figure 3